NightSafe Routes

Fear-Free Night Navigator · The Big Code 2026 Hackathon

Participant Name	[Add your full name]
Email ID	[Add the email ID used for hackathon communication]
Year of Degree	[1st / 2nd / 3rd / 4th / Dual Degree year]
GitHub Repository	https://github.com/[your-username]/nightsafe-routes
README	https://github.com/[your-username]/nightsafe-routes#readme

BRIEF SUMMARY

NightSafe Routes is an AI-assisted navigation prototype for Bangalore that treats **psychological safety** as a first-class routing objective for night travel. Rather than optimising purely for ETA, the system assigns each road segment a *Safety Score* derived from geospatial proxy signals — lighting intensity, commercial activity, street connectivity, transit proximity, and dead-end risk — then exposes four route variants: **Fastest, Balanced, Safest,** and a **personalised Agent Route**. The prototype integrates a FastAPI backend, a React + Leaflet map interface, SHAP-based segment-level explanations, and a Pareto-frontier visualisation of ETA vs. safety trade-offs.

PROBLEM STATEMENT

Standard navigation apps optimise almost exclusively for minimum travel time, frequently routing users through dark, isolated, or low-activity streets to save a few minutes. For solo travellers and women commuting at night, the shortest path can be deeply anxiety-inducing — and may drive costly behavioural changes such as avoiding trips or paying for taxis unnecessarily.

NightSafe Routes addresses the **Fear-Free Night Navigator** challenge: generating routes that are not merely shorter but *safer to traverse under nighttime conditions*, prioritising peace of mind for solo commuters, women riders, students, and other vulnerable road users.

DESIGN IDEA AND APPROACH

Pipeline Overview

- Load a Bangalore road graph and POI data from OpenStreetMap (OSMnx).
- Engineer edge-level safety proxy features (lighting, activity, connectivity, transit proximity, dead-end penalty).
- Generate synthetic training labels that vary by time-of-day (default / evening / night).
- Train three XGBoost regressors to predict time-aware safety scores per segment.
- Train an MLP archetype classifier to map user context (travel mode, hour, gender, destination type) to one of four routing archetypes.
- Train per-archetype route agents with PPO (Stable-Baselines3); fall back to Q-learning when throughput is insufficient.
- Serve route variants and SHAP-derived segment explanations via a FastAPI + React frontend.

Technologies

- **Backend:** Python · FastAPI · Uvicorn · OSMnx · NetworkX · GeoPandas · Shapely
- **ML / RL:** XGBoost · Scikit-learn MLP · Stable-Baselines3 PPO · Q-learning · SHAP
- **Frontend:** React 18 · Vite · Leaflet · Recharts

Scaling & Rollout

The dominant scaling parameters are the number of road segments, concurrent route requests, and RL agent training time over large subgraphs. The design pre-computes and caches processed graphs and trained models, serving routes from an in-memory graph to keep per-request latency low (mean 414 ms, p95 730 ms). A staged rollout begins with a single Bangalore corridor at night-only hours, collects user comfort feedback, then progressively adds richer real-world supervision (lighting surveys, incident reports) before expanding to further neighbourhoods and cities.

Privacy & Security

The prototype uses lightweight, ephemeral user context (travel mode, hour, destination type, female-rider flag). A production deployment would minimise retained personal data, avoid storing trip histories by default, and enforce strict consent and deletion policies for any sensitive attributes.

PROTOTYPE SNAPSHOT

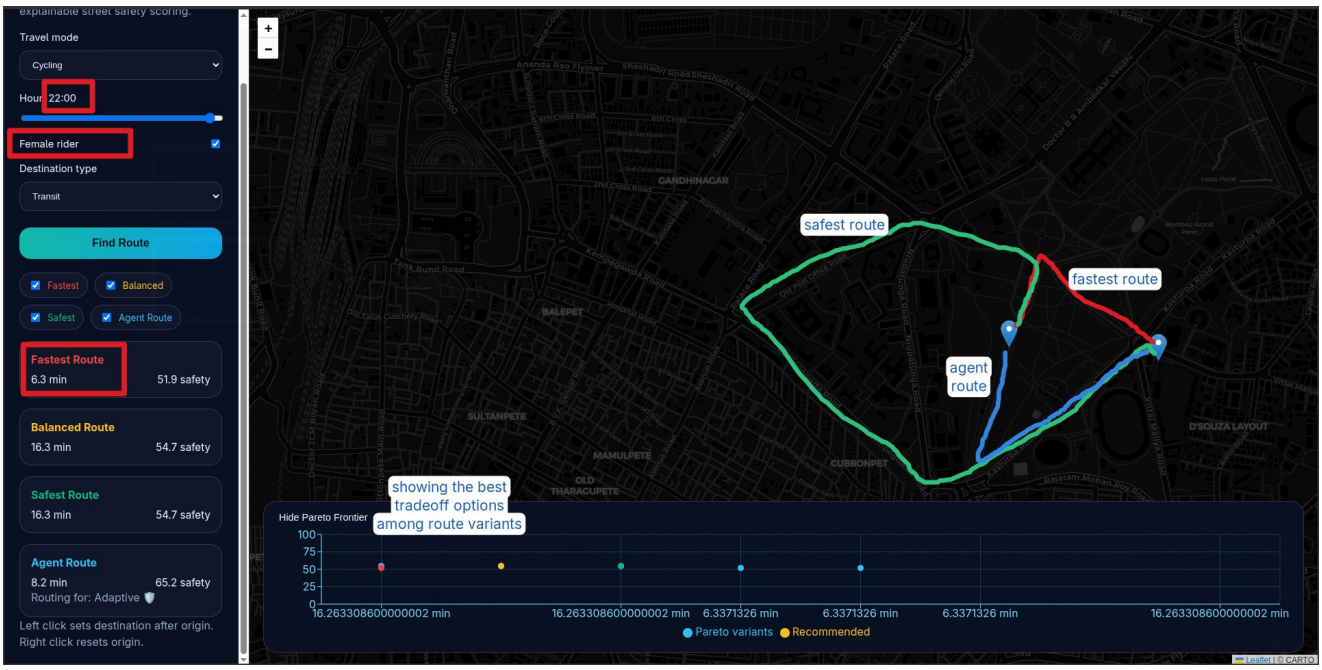

Figure 1 — NightSafe Routes UI showing four route variants on a dark Bangalore basemap, route summary cards (ETA + mean safety score), and the Pareto-frontier scatter plot.

IMPACT

Many people — especially women and solo travellers — change behaviour, spend extra money on cabs, or avoid trips altogether because standard maps ignore night-time safety concerns. NightSafe Routes addresses this gap by internalising psychological safety as a measurable routing objective.

- Safer recommended paths for vulnerable users, grounded in concrete urban signals.
- Higher confidence in night travel through explicit ETA vs. safety trade-off control.
- Improved transparency via SHAP-derived segment explanations (e.g. 'isolated dead-end segment' or 'near transit stops').
- Modular architecture allowing progressive improvement as stronger real-world labels (incident data, lighting surveys) become available.

FEASIBILITY

The implementation is feasible because a working vertical slice is already in place:

- Processed geospatial graph artifacts, trained XGBoost models, RL agents, and SHAP explanations are all versioned in the repository.
- Backend and frontend components are integrated and tested with automated test suites covering the API, routing logic, pipeline, and RL environment.

- The synthetic label strategy — using high-quality proxy signals when real-world incident data is unavailable — is explicitly encouraged in the problem statement and produces credible scores.
- The modular design allows stronger supervision to be swapped in later without redesigning the system.

USE OF AI

AI is central to every layer of the solution, not merely decorative:

- **XGBoost regression** — infers edge-level safety from seven geospatial proxy features with Optuna-tuned hyperparameters.
- **SHAP explainability** — provides segment-level feature attributions rendered directly in the frontend.
- **MLP archetype classifier** — maps user context to one of four routing archetypes (Vulnerable Solo, Comfort Seeker, Efficiency First, Adaptive).
- **PPO reinforcement learning** (with Q-learning fallback) — learns archetype-specific route behaviour from an OpenAI Gym-compatible environment.

DEMONSTRABLE RELIABILITY

Safety Scoring Models

Model	RMSE	MAE
Default (day)	3.08	2.46
Evening model	3.03	2.42
Night model	3.01	2.39

- **Archetype classifier accuracy:** 59.9 %
- **Full-model mean safest-route safety:** 76.57 / 100
- **Route latency — mean / p95 / p99:** 414 ms / 730 ms / 989 ms

Feature-importance rankings consistently highlight **dead-end penalty** and **activity score** as dominant signals. Ablation studies confirm that removing activity score, connectivity score, or time-aware scoring each degrades safest-route mean safety — validating the design assumptions. Notably, the personalised agent route currently underperforms the heuristic safest route in offline evaluation; this honest result surfaces a clear next step — improved reward shaping and extended policy training — rather than masking it.

ALTERNATIVES CONSIDERED

- **Rule-only scoring** — Simpler but too brittle and hard to calibrate across different neighbourhood types.
- **End-to-end deep route recommendation** — Less interpretable and unrealistic given the limited labelled trajectory data available.
- **Crime-data-only routing** — Attractive in theory, but crime datasets are often sparse, delayed, and poorly aligned to block-level perception.
- **Purely personalised routing without a shared safety model** — Weaker transparency and harder to justify in high-trust mobility settings.

REFERENCES & APPENDICES

- **OpenStreetMap** — road network and POI data (downloaded via OSMnx).
- **Repository artifacts:** *data/evaluation_report.json* (offline metrics), *data/processed/training_curves.json* (RL training snapshots), *data/processed/shap_explanations.json* (per-segment SHAP values).
- **API:** FastAPI endpoints — *POST /route* (route variants), *GET /segment/{edge_id}/explain* (SHAP explanation), *GET /health* (graph stats).
- **Frontend:** React 18 + Leaflet — dark basemap, four overlaid colour-coded route polylines, segment tooltip on hover, Pareto scatter chart.

- **Tests:** *tests/test_api.py, test_classifier.py, test_pipeline.py, test_rl_env.py, test_router.py* — covering all major subsystems.

Before final submission, replace the participant placeholders in the metadata table and insert the public GitHub repository link.